STAT201 · STATISTICS · LECTURE 5

Continuous Distributions and the Normal Distribution

Density, area under a curve, standardization and working with the normal table

Syllabus topic 5 · week 5 · 4 academic hours

LECTURE GOALS

- explain why for a continuous variable probability is the area under the density and $P(X = x) = 0$;
- work with the uniform and exponential distributions;
- standardize a normal variable (z-score) and find probabilities and quantiles from the table;
- apply the 68–95–99.7 rule and find unknown parameters from a given probability;
- use the normality of a sum of independent normal variables and the normal approximation to the binomial distribution.

1 Probability density

For a continuous variable (time, mass, an amount of money) one cannot list the values and assign probabilities to them: there are infinitely many values, and the probability of any single value is zero. Instead a *density* $f(x)$ is specified.

PROBABILITY DENSITY

A function $f(x) \geq 0$ with total area under the curve equal to 1 is called a density. The probability of falling into an interval is the area under the curve above it:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad P(X = x) = 0. \tag{1.1}$$

Therefore for a continuous variable it does not matter whether inequalities are strict or non-strict: $P(X < b) = P(X \leq b)$. The distribution function is defined as before: $F(x) = P(X \leq x)$ (the area to the left of x), and $P(a < X < b) = F(b) - F(a)$. The expectation and variance are given by the integrals $\mu = \int x f(x) dx$, $\sigma^2 = \int (x-\mu)^2 f(x) dx$, but for the standard families we shall use ready-made formulas.

COMMON MISTAKE

The value of the density $f(x)$ is not a probability and may exceed 1 (for example, the density of the uniform distribution on $[0, 0.5]$ equals 2). Probability is only area.

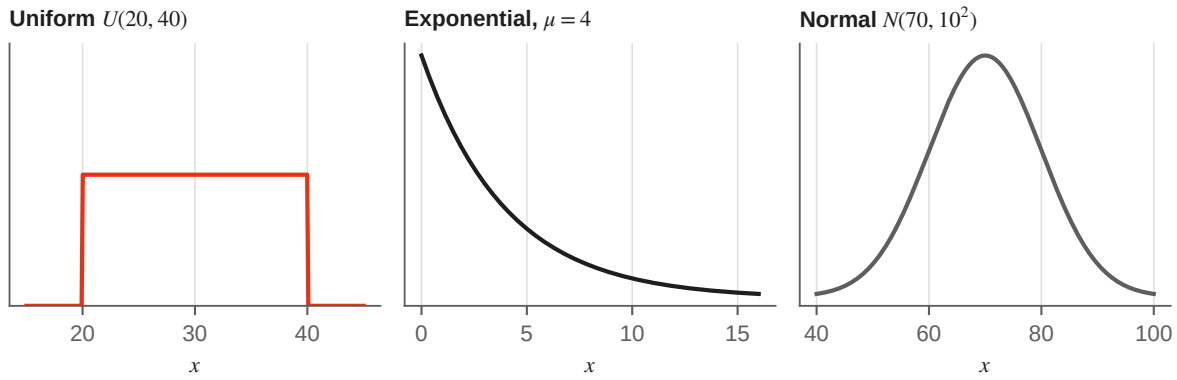


Fig. 1. Three densities: uniform (delivery time, min), exponential (waiting, min), normal (scores).

2 The uniform distribution

If all values on the interval $[a, b]$ are equally likely, $X \sim U(a, b)$:

$$f(x) = \frac{1}{b-a}\ (a \le x \le b), \qquad P(c \le X \le d) = \frac{d-c}{b-a}, \qquad \mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}. \tag{2.1}$$

EXAMPLE 1. DELIVERY TIME

A courier delivers an order in time $X \sim U(20, 40)$ minutes. $\mu = 30$, $\sigma = 20/\sqrt{12} = 5.77$. $P(25 \le X \le 35) = 10/20 = 0.5$; $P(X > 36) = 4/20 = 0.2$.

3 The exponential distribution

The waiting time until the next event of a Poisson flow (Lecture 4) with intensity λ is exponentially distributed:

$$f(x) = \lambda e^{-\lambda x}, \quad P(X > x) = e^{-\lambda x}, \quad F(x) = 1 - e^{-\lambda x}\ (x \ge 0); \qquad \mu = \sigma = \frac{1}{\lambda}. \tag{3.1}$$

EXAMPLE 2. WAITING IN A QUEUE

The average wait for service is 4 minutes, $\lambda = 1/4$. $P(X > 6) = e^{-1.5} = 0.2231$; $P(X < 2) = 1 - e^{-0.5} = 0.3935$; $P(3 < X < 5) = e^{-0.75} - e^{-1.25} = 0.4724 - 0.2865 = 0.1859$. The median $= 4 \ln 2 = 2.77$ minutes, less than the mean of 4: the distribution is right-skewed.

REMEMBER

The exponential distribution has no “memory”: $P(X > s + t \mid X > s) = P(X > t)$. If you have already waited 5 minutes, the remaining waiting time is distributed the same as for a customer who has just arrived.

4 The normal distribution

NORMAL DISTRIBUTION

$X \sim N(\mu, \sigma^2)$ if the density has the bell shape

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{4.1}$$

The curve is symmetric about μ; the parameter σ determines the width.

Variables that are formed as a sum of many small independent influences are (approximately) normal: measurement errors, height, scores on large tests, and, as we shall see in Lecture 7, sample means for any initial distribution.

4.1 Standardization

The area under $N(\mu, \sigma^2)$ cannot be taken "head-on", so all probabilities are reduced to the standard normal $Z \sim N(0, 1)$ using the z-score:

$$z = \frac{x - \mu}{\sigma}, \qquad P(X \le x) = \Phi(z), \tag{4.2}$$

where $\Phi(z)$ is the distribution function of Z (table in the appendix "Statistical Tables", Excel: `NORM.S.DIST(z,TRUE)`). The value z shows by how many standard deviations a value lies from the mean.

Useful properties: $\Phi(-z) = 1 - \Phi(z)$; $P(Z > z) = 1 - \Phi(z)$; $P(a < Z < b) = \Phi(b) - \Phi(a)$.

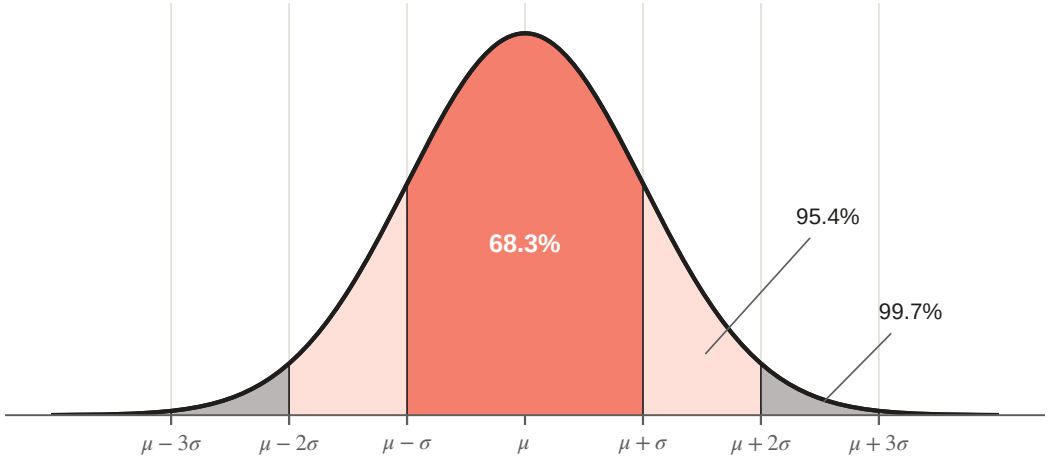


Fig. 2. The 68–95–99.7 rule: for a normal variable 68.3% of values lie within $\pm 1\sigma$, 95.4% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

EXAMPLE 3. EXAM SCORES

Scores $X \sim N(70, 10^2)$.

- $P(X > 85)$: $z = 1.5$, $P = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$.
- $P(60 < X < 80)$: $z = \pm 1$, $P = 0.8413 - 0.1587 = 0.6827$.
- $P(65 < X < 90)$: $z = -0.5$ and 2: $P = 0.9772 - 0.3085 = 0.6687$.

4.2 The inverse problem: quantiles

To find the value x above (below) which a given share lies, we find z from the table "in reverse" and go back: $x = \mu + z\sigma$. Frequently used values: $z_{0.90} = 1.2816$, $z_{0.95} = 1.6449$, $z_{0.975} = 1.9600$, $z_{0.99} = 2.3263$, $z_{0.995} = 2.5758$.

EXAMPLE 4. THE TOP 10% AND THE QUARTILES

The cut-off for the top 10% in the exam: $x = 70 + 1.2816 \cdot 10 = 82.8$. The lower quartile is $Q_1 = 70 - 0.6745 \cdot 10 = 63.3$, the upper $Q_3 = 76.7$; hence $IQR = 13.5 = 1.349\,\sigma$ (a useful guide from Lecture 1: for normal data $IQR \approx 1.35\sigma$).

EXAMPLE 5. INCOMES

Monthly income $X \sim N(5.5;\ 1.2^2)$ million UZS. $P(X > 7) = P(Z > 1.25) = 0.1056$; $P(X < 4) = 0.1056$ (symmetry: $z = -1.25$); $P(4 < X < 7) = 0.7887$. The central 95% of values: $5.5 \pm 1.96 \cdot 1.2$, i.e. from 3.15 to 7.85 million.

4.3 Unknown parameters

If the probability is known but a parameter is not, we write down the standardization and solve the equation. For example, $X \sim N(90, \sigma^2)$ and $P(X > 100) = 0.10$. Then $z = 1.2816 = \dfrac{100 - 90}{\sigma}$, $\sigma = 10/1.2816 = 7.80$. Another example: $X \sim N(\mu, 8^2)$ and $P(X < 50) = 0.20$: $z_{0.20} = -0.8416$, $50 = \mu - 0.8416 \cdot 8$, $\mu = 56.73$.

4.4 Comparison through z-scores

The z-score allows results on different scales to be compared. A student scored 78 on a test $N(70, 8^2)$ and 640 on a test $N(580, 60^2)$: $z_1 = 1.0$, $z_2 = 1.0$. The relative results are the same.

4.5 Sums of normal variables

A linear combination of independent normal variables is normal, and the parameters are computed by the rules of Lecture 4. Let $X \sim N(40, 6^2)$, $Y \sim N(25, 8^2)$ be independent. Then $X + Y \sim N(65, 10^2)$ and $X - Y \sim N(15, 10^2)$; $P(X + Y > 80) = P(Z > 1.5) = 0.0668$, $P(X < Y) = P(X - Y < 0) = P(Z < -1.5) = 0.0668$.

5 Normal approximation to the binomial

For large n the distribution $\mathrm{Bin}(n, p)$ is well approximated by the normal $N(np,\ np(1-p))$ (condition: $np \geq 10$ and $n(1-p) \geq 10$). Since the binomial distribution is discrete, a *continuity correction* is introduced: $P(X \leq k) \approx \Phi\left(\dfrac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$.

EXAMPLE 6. 100 CUSTOMERS

$X \sim \text{Bin}(100, 0.3)$: $\mu = 30$, $\sigma = \sqrt{21} = 4.58$. Exactly $P(X \le 35) = 0.8839$. Without the correction $\Phi(1.09) = 0.8624$, with the correction $\Phi\left(\frac{35.5-30}{4.58}\right) = \Phi(1.20) = 0.8849$. The correction noticeably improves the accuracy of the approximation.

6 Key terms

Term	*In Russian*	Meaning
Probability density function (pdf)	*Плотность*	$P(a \le X \le b) = \int_a^b f$
Uniform	*Равномерное*	$\mu = \frac{a+b}{2}$, $\sigma^2 = \frac{(b-a)^2}{12}$
Exponential	*Экспоненциальное*	$P(X > x) = e^{-\lambda x}$, $\mu = 1/\lambda$
Normal	*Нормальное*	bell-shaped, $N(\mu, \sigma^2)$
z-score	*z-оценка*	$z = (x - \mu)/\sigma$
Standard normal	*Стандартное нормальное*	$Z \sim N(0, 1)$, function Φ
Quantile	*Квантиль*	the value to the left of which lies a given share
Continuity correction	*Поправка на непрерывность*	± 0.5 when approximating a discrete variable

Formulas of Lecture 5

$$U(a,b):\ \mu = \tfrac{a+b}{2},\ \sigma^2 = \tfrac{(b-a)^2}{12}; \qquad \text{Exp}(\lambda):\ P(X > x) = e^{-\lambda x},\ \mu = \sigma = \tfrac{1}{\lambda};$$

$$X \sim N(\mu, \sigma^2):\ z = \frac{x - \mu}{\sigma},\ \ P(X \le x) = \Phi(z),\ \ x = \mu + z\sigma;$$

$$\text{Bin}(n,p) \approx N(np,\ np(1-p)),\ \ P(X \le k) \approx \Phi\Big(\tfrac{k+0.5-np}{\sqrt{np(1-p)}}\Big).$$

7 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] A bus runs at 15-minute intervals. A passenger arrives at a random moment, the waiting time is $X \sim U(0, 15)$.

(a) Find μ and σ.
(b) Find $P(X > 10)$ and $P(5 < X < 8)$.

Problem 2 [A] The service time of a customer at a bank is exponential with a mean of 5 minutes.

(a) What is the parameter λ?
(b) Find the probability that service lasts more than 3 minutes, and the probability that it lasts less than 10 minutes.
(c) Find the median.

Problem 3 [A] $X \sim N(200, 20^2)$. Find $P(X < 230)$, $P(X > 170)$, $P(190 < X < 215)$.

Problem 4 [B] A machine packs coffee, the mass of a pack is $X \sim N(500, 4^2)$ g.

(a) Find the share of packs weighing less than 495 g.
(b) Find the share of packs weighing more than 510 g.
(c) To what mean should the machine be set so that only 1% of packs weigh less than 495 g (the standard deviation stays the same)?

Problem 5 [B] The size of a consumer loan is $X \sim N(40, 12^2)$ million UZS.

(a) Find the share of loans larger than 60 million.
(b) Find the share of loans smaller than 30 million.
(c) Find the cut-off for the largest 5% of loans and the interval containing the central 90% of loans.

Problem 6 [B] For $X \sim N(90, \sigma^2)$ it is known that $P(X > 100) = 0.10$. For $Y \sim N(\mu, 8^2)$ it is known that $P(Y < 50) = 0.20$. Find σ and μ.

Problem 7 [B] Student A scored 82 on an exam $N(70, 8^2)$, student B scored 640 on an exam $N(580, 60^2)$. Who showed the better relative result? What share of test-takers scored lower than each?

Problem 8 [B] The returns of two independent assets (%): $R_1 \sim N(8, 4^2)$, $R_2 \sim N(5, 3^2)$. The portfolio is the sum $R_1 + R_2$.

(a) Find the distribution of $R_1 + R_2$.
(b) Find the probability of a negative total return.
(c) Find the probability that $R_1 < R_2$.

Problem 9 [B] 40% of a shop's visitors make a purchase, independently. 200 visitors came in a day.

(a) Check that the normal approximation is applicable and find μ and σ of the number of buyers.
(b) Find the probability that there are at least 90 buyers (with a continuity correction).

Problem 10 [B] The time between customer arrivals is exponential with a mean of 2 minutes.

(a) Find the probability of waiting longer than 3 minutes for the next customer.
(b) Two minutes have already passed with no customers. What is the probability that we must wait at least 3 more minutes?
(c) What is the distribution of the number of customers in 10 minutes? Find the probability that nobody arrives in 10 minutes.

Problem 11 [C] Prove the memorylessness of the exponential distribution: $P(X > s + t \mid X > s) = P(X > t)$.

Problem 12 [C] For $X \sim U(a, b)$ derive $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ by integration.

Problem 13 [C] Let $X \sim N(\mu, \sigma^2)$. Show that the z-score $Z = (X - \mu)/\sigma$ has $\mathrm{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. Explain why $IQR \approx 1.349\sigma$ implies that for $IQR = 27$ in normal data $\sigma \approx 20$.